

The Smallest GPS TRACKER Ever

ASHWINI KUMAR SINHA

Global positioning system (GPS)-based tracking is one of the most important technologies in today's world, which

is very useful from parcel updates to navigation and many different fields of human life. However, if one wants to track small items,

baggage, children's locations, or spy on someone, a very small and compact-sized tracker is often needed, which can easily be at-

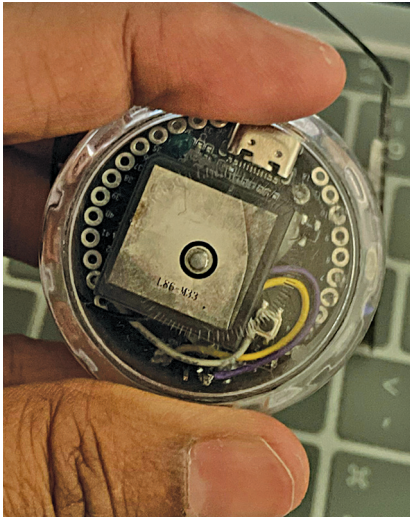


Fig. 1: The tiny GPS tracker

Features	Implementation
GNSS	GPS + GLONASS
Power Supply	Supply voltage: 2.8 V to 4.3 V, typ. 3.3 V
Power Consumption	- Acquisition: 26 mA @ VCC = V_BCKP = 3.3 V (GPS) - Tracking: 22 mA @ VCC = V_BCKP = 3.3 V (GPS) - Acquisition: 30 mA @ VCC = V_BCKP = 3.3 V (GPS+GLONASS) - Tracking: 26 mA @ VCC = V_BCKP = 3.3 V (GPS+GLONASS) - Standby: 1 mA @ VCC = V_BCKP = 3.3 V - Backup: 7 μA @ V_BCKP = 3.3 V
Receiver Type	- GPS L1 1575.42 MHz C/A Code - GLONASS L1 1598.0625–1605.375 MHz C/A Code
Sensitivity	- Acquisition: -149 dBm - Re-acquisition: -161 dBm - Tracking: -167 dBm
TTFF (EASY enabled)	- Cold start: typ. 15 s @ -130 dBm - Warm start: typ. 5 s @ -130 dBm

Fig. 3: GPS powering system details (Source: Datasheet of L86 GPS module)

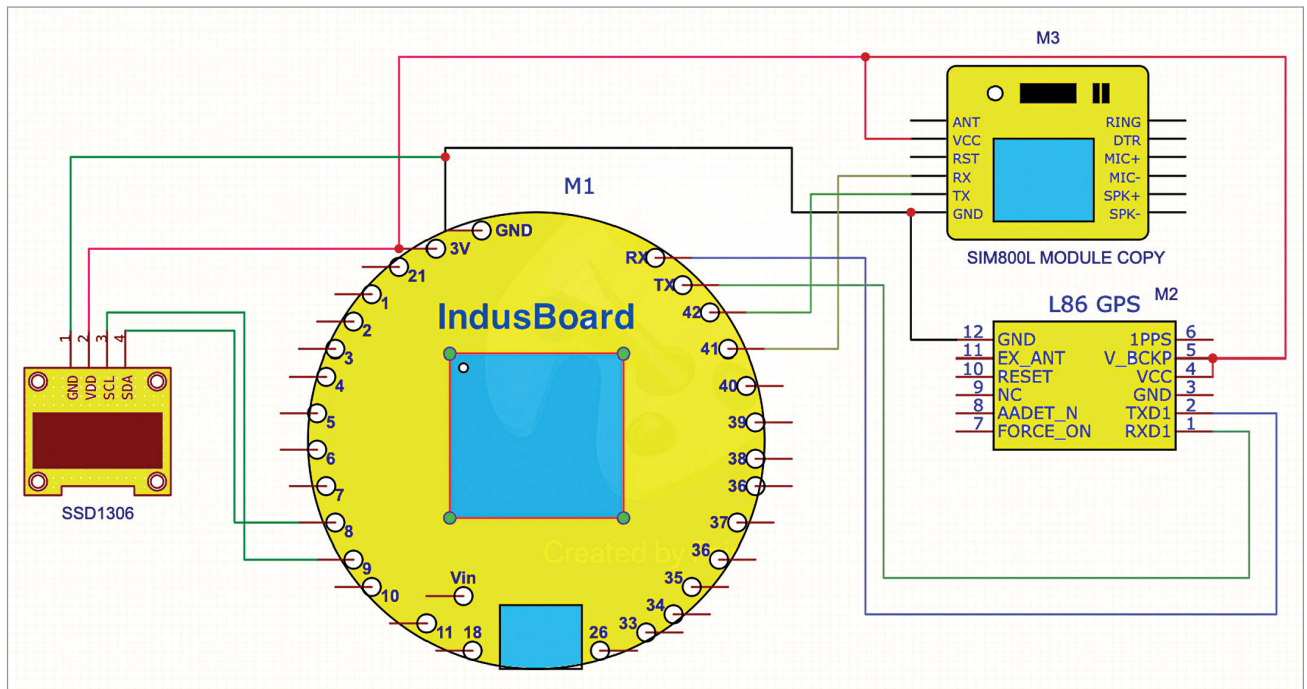


Fig. 2: Connections for the tiny GPS tracker device